BUILD AND EVALUATE A LINEAR REGRESSION MODEL

California Housing Price Prediction

Internship Organization: Maincrafts Technology
Internship Domain: Artificial Intelligence & Machine Learning (AI/ML)
Task: Task 1 – Linear Regression Model
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1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions. This project focuses on developing a Linear Regression model to predict median house values using the California Housing dataset.

The project introduces the fundamental machine learning workflow, including data loading, exploratory data analysis, preprocessing, model training, prediction, evaluation and reporting.

2. Objectives

- Understand the basic machine learning workflow.
- Explore and analyse the California Housing dataset.
- Develop a Linear Regression prediction model.
- Evaluate model performance using MAE, RMSE and R^2.
- Visualise actual and predicted values and analyse residuals.

3. Dataset Description

The California Housing dataset was obtained using the fetch_california_housing() function from Scikit-learn.

Dataset details:

- Total observations: 20,640
- Input features: 8
- Target variable: MedHouseVal
- Missing values: 0
- Duplicate records: 0

The input variables include median income, house age, average rooms, average bedrooms, population, average occupancy, latitude and longitude.

The target variable, MedHouseVal, represents median house value in units of $100,000.

4. Tools and Technologies

- Python 3.11
- NumPy and Pandas
- Scikit-learn

- Matplotlib
- Seaborn
- Jupyter Notebook
- Visual Studio Code

5. Methodology

The project followed a standard supervised machine learning workflow.

5.1 Data Loading and Exploration

The California Housing dataset was loaded using Scikit-learn. Pandas was used to inspect the dataset structure, summary statistics, missing values and duplicate records.

5.2 Exploratory Data Analysis

Exploratory data analysis was performed to understand the distribution of the target variable and relationships among the input features. A histogram and correlation heatmap were generated.

5.3 Data Splitting

The dataset was divided into training and testing sets using an 80:20 ratio. The training set contained 16,512 observations, while the testing set contained 4,128 observations. A random state of 42 was used for reproducibility.

5.4 Model Training

A Linear Regression model from Scikit-learn was trained using the eight input features. The trained model was then used to predict house values for the test dataset.

5.5 Model Evaluation

The model was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2).

6. Results and Model Evaluation

The Linear Regression model produced the following evaluation results:

Evaluation Metric	Result
Mean Absolute Error (MAE)	0.533
Root Mean Squared Error (RMSE)	0.746
R^2 Score	0.576

Interpretation:

- The MAE of 0.533 indicates an average absolute prediction error of approximately $53,300.
- The RMSE of 0.746 indicates the model's prediction error, with greater weight given to larger errors.
- The R^2 score of 0.576 indicates that the model explains approximately 57.6% of the variation in median house values in the test dataset.

Actual-versus-predicted and residual plots were generated to visually assess model performance.

7. Conclusion

A Linear Regression model was successfully developed and evaluated using the California Housing dataset. The project provided practical experience in data loading, exploratory data analysis, data splitting, model training, prediction and performance evaluation.

The model achieved an R^2 score of 0.576. The results demonstrate the usefulness of Linear Regression as a baseline prediction model. Further improvements could be explored using feature engineering, data transformation and more advanced machine learning algorithms.

8. Project Deliverables

- Jupyter Notebook: task1_ml_linear_regression.ipynb
- Saved model: house_price_model.pkl
- GitHub Repository: https://github.com/neelsongeorgedawson-star/ML_Task1_California_Housing

9. Acknowledgement

I sincerely thank Maincrafts Technology for providing this practical learning opportunity in Artificial Intelligence and Machine Learning. This task helped me strengthen my understanding of the machine learning workflow and gain hands-on experience with Python and Scikit-learn.

10. Visualisation and Graphical Results

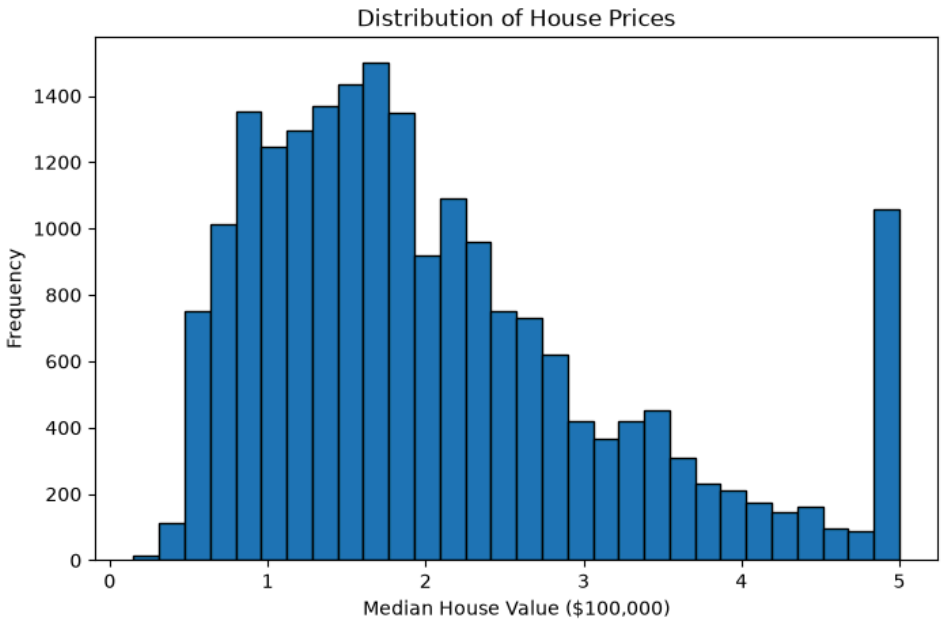


Figure 1. Distribution of median house values in the California Housing dataset.

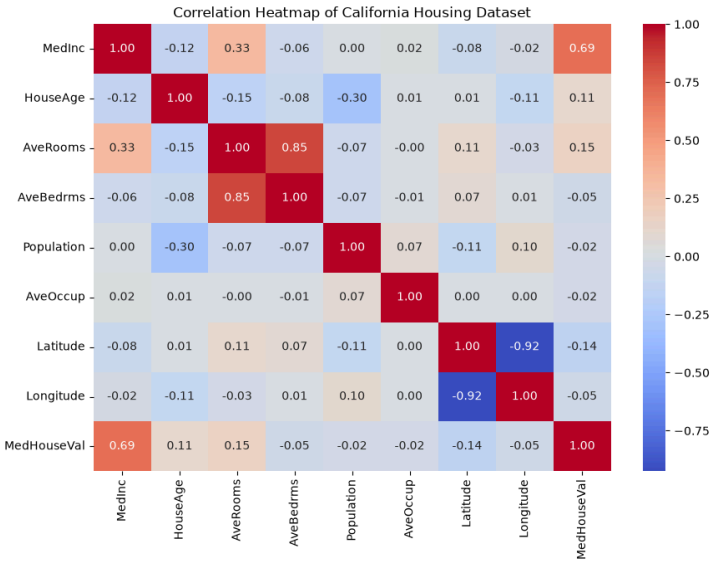


Figure 2. Correlation among the housing dataset variables.

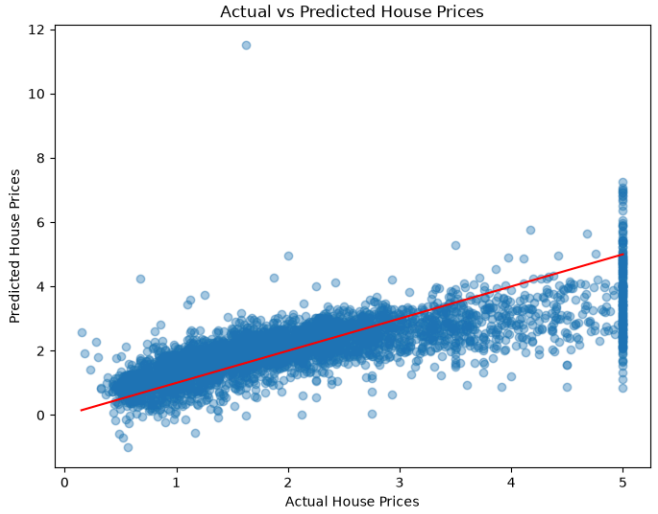


Figure 3. Actual versus predicted median house values.

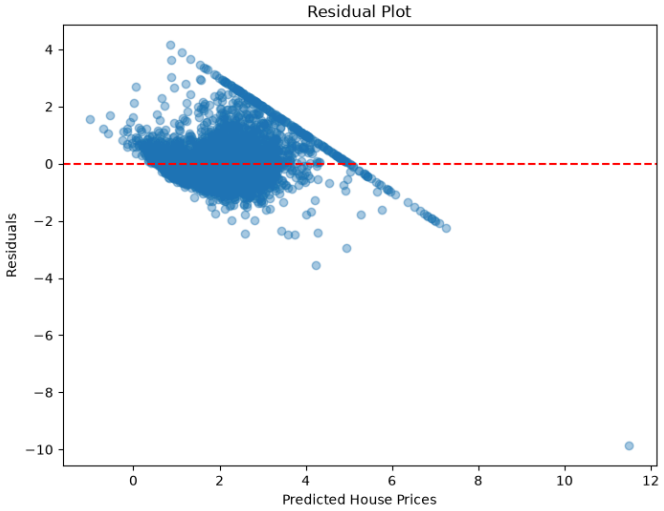


Figure 4. Residual plot of the Linear Regression model.